Sophia Sun

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Education

University of California San Diego, San Diego, CA — Ph.D. 2018 - 2025
(spatiotemporal machine learning, uncertainty quantification, decision-making under uncertainty)
Artificial Intelligence Group, Computer Science and Engineering
Advisor: Prof. Rose Yu

Wellesley College, Wellesley, MA — B.A. 2014 – 2018
Double Major in Mathematics and Computer Science.

Massachusetts Institute of Technology, Cambridge, MA — *Cross Registration* 2016 – 2018
Coursework in Mathematics and Computer Science

Experiences

Applied Scientist, AWS AI Fundamental Research Santa Clara, CA 2025 – present
(Agentic Systems, RAG, LLM Post-training and evaluation)
* AI agents for amazon Quick. Shipped the web agent and helped push the desktop app from conception to production. I worked on the agent harness optimization (fine tuning of LMs, routing), context layer (knowledge graph, memory, RAG, and search tooling), built the evaluation framework (eval harness + benchmarks used by 50+ researchers), and skills for artifact generation.

Applied Scientist Intern, AWS AI Santa Clara, CA 2023, 2024
(time-series forecasting, uncertainty quantification, anomaly detection)
* Designed unsupervised **anomaly detection** algorithm for AWS database usage through conformal prediction and quantile tracking methods; eliminating cold start in the RDS anomaly detection pipeline.
(ambiguity detection, memory in LLM agents)
* Designed and implemented pipeline to handle **ambiguities in the LLM**-powered AWS Text-to-SQL product.

Research Scientist Intern, Amazon Robotics RAD 2020
(reinforcement learning, multi-agent planning, optimization)
* Machine learning for multi-robot trajectory planning in amazon's fulfillment centers.
* Help built simulation for multi-robot trajectory planning and improved overall throughput by 3% in simulation.

Engineering Intern, Google DeepMind London, U.K. 2018
(backend software engineering, data science)
* Secure Electronic Health Records data pipelining, analysis, and verification.
* Improved interpretability and verifiability of analytical algorithms deployed in Deepmind's Streams App.

Research Assistant, Johns Hopkins University Baltimore, MD 2017 - 2018
(machine learning for healthcare, data science)
PI: Prof. Suchi Saria; NSF Research Experience for Undergraduate program
* Predicting in-Hospital Cardiac Deterioration from Heterogeneous multi-resolution medical time-series data.

Research Assistant, MIT http://web.mit.edu/music21/ Cambridge, MA 2016 - 2018
(machine learning for music, digital humanities)
PI: Prof. Michael Cuthbert
* Investigated the expression of sentiments in songs based on lyrics, melodic, harmonic, and other musical data extracted from sheet music through computational methods. (Paper in Empirical Musicology Review)

Publications

Machine Learning and Uncertainty Quantification

Sophia Sun. "Probabilistic Deep Learning for Uncertainty Quantification and Decision-Making." *Doctoral dissertation, University of California, San Diego*. 2025

Sophia Sun, Rose Yu. "Conformal Prediction for Time-series Forecasting with Change Points." In proceedings of *Neural Information Processing systems. NeurIPS 2025*

Sophia Sun, Abishek Sankararaman, Murali Narayanaswamy. "Online Adaptive Anomaly Thresholding with Confidence Sequences." *International Conference on Machine Learning, ICML* 2024

Sophia Sun, Rose Yu. "Copula Conformal Prediction for Multi-step Time Series Forecasting." *International Conference on Learning Representations ICLR*, 2024

Sophia Sun, Wenyan Chen, Zihao Zhou, Sonia Fereidooni, Elise Jortberg, Rose Yu. "Data-Driven Simulator for Mechanical Circulatory Support with Domain Adversarial Neural Process." *Learning for Dynamics and Control Conference (L4DC),* 2024

Sophia Sun, Robin Walters, Jinxi Li, Rose Yu. "Probabilistic Symmetry for Multi-Agent Dynamics." *Learning for Dynamics and Control Conference (L4DC),* 2023

Large Language Models

Yiyou Sun, Xinyang Han, et. al. (incl. **Sophia Sun** as contributing author) "Agents' Last Exam" preprint, 2026

Sophia Sun, Abishek Sankararaman, Murali Narayanaswamy. "Resolving Ambiguity through Personalization in LLM chat systems." Workshop on Reasoning and Planning for Large Language Models @ *ICML* 2025

Bodhisattwa Prasad Majumder, Shuyang Li, Jianmo Ni, Huanru Henry Mao, **Sophia Sun**, Julian McAuley. "Bernard: A Stateful Neural Open-domain Socialbot." *Alexa Prize Proceedings*, 2020

Reinforcement Learning

Aysin Tumay*, **Sophia Sun***, Sonia Fereidooni, Aaron Dumas, Rose Yu. "Uncertainty-Aware Offline Reinforcement Learning for Safe Medical Decisions." *In submission*

Sophia Sun*, Aysin Tumay*, Sonia Fereidooni, Aaron Dumas, Rose Yu. "Designing Digital Twin to Support Safe Medical Decision-Making." *In submission*

Sander Tonkens*, **Sophia Sun***, Rose Yu, Sylvia Herbert. "Scalable Safe Long-Horizon Planning in Dynamic Environments Leveraging Conformal Prediction and Temporal Correlations.*" IEEE International Conference on Robotics and Automation workshop on Long-term Human Motion Prediction (ICRA)*, 2023

Creative AI and Digital Humanities

Cheng, Mingyong, **Sophia Sun**, Han Zhang, and Yuemeng Gu. "Learning to Move, Learning to Play, Learning to Animate: a Multimedia Exploration of the More-than-human Intelligence.*" Proceedings of the ACM on Computer Graphics and Interactive Techniques 8, no. 3: 1-11 (SIGGRAPH 2025)*

Sofy Yuditskaya*, **Sophia Sun***, and Margaret Schedel*. "Synthetic Erudition Assist Lattice." *New Interfaces for Musical Expression* (*NIME), 2021*

Sofy Yudistkaya*, Derek Kwan*, **Sophia Sun***. "Karaoke of Dreams: A multi-modal neural-network generated music experience." *Proceedings of the Joint Conference on AI Music Creativity (CIMC+MUME),* 2020

Sophia Sun, Michael Scott Cuthbert. "Emotion painting: lyric, affect, and musical relationships in a large lead-sheet corpus." *Empirical Musicology Review 12.3-4*. 2018

Service, Talks, and Workshops

Reviewer for ICML 2022 - 2025, NeurIPS 2022 - 2025, ICLR 2023 - 2025, JMLR 2025

Learning to Move, Learning to Play, Learning to Animate Contributed Talk at GenAI Summit 2025	San Diego, CA	2024
Workshop on Statistical Frontiers in LLMs and Foundation Models NeurIPS Workshop Program Committee	Vancouver, BC	2024
Conformal Methods for Quantifying Uncertainty for Time-series Data Talk at AWS AI	Sunnyvale, CA	2023
Conformal Prediction for Spatio-Temporal Uncertainty Quantification	Sunnyvale, CA	2023

Talk at UCSD CSE

Probabilistic Symmetry for Improved Trajectory Forecasting San Diego, CA 2022
Talk at UCSD CSE Research Open House AI spotlight

Reinforcement Learning 101 online 2020
Three-part workshop series for artists at Arts Counsel Korea (ARKO)

Natural Language Processing for Creatives Bombay Beach, CA 2020
Workshop at mars.college residency

Teaching

Lecturer & head TA, CSE251b Graduate level Deep Learning, UC San Diego Spring 2025
Lecturer & head TA, CSE151b Deep Learning, UC San Diego Spring 2023
Head TA, CSE 291 Deep Generative Models, UC San Diego Fall 2022
Teaching Assistant, CSE101 Design & Analysis of Algorithms, UC San Diego Summer 2022
Teaching Assistant, CSE20 Discrete Mathematics, UC San Diego Summer 2022
Teaching Assistant, CSE251C Machine Learning Theory, UC San Diego Spring 2021
Teaching Assistant, CSE291 Topics in Search and Optimization, UC San Diego Winter 2020

Leadership

Women in Data Science (WiDS), UC San Diego Organizer, 2023 – 2024
UCSD ACM-W Student Chapter, UC San Diego Chair, 2020 – 2022
Graduate Women in Computing (GradWIC), UC San Diego Vice President, 2019 – 2022

Skills

AI systems: Agentic systems, retrieval augmented generation (RAG), harness optimization and model routing, LLM post-training, evaluation of LLM and Agentic systems.

Machine Learning: generative models, probabilistic spatial-temporal forecasting, world models/digital twin learning, anomaly detection, Transformers and LLMs.

Reinforcement Learning: Designing RL pipeline for real-world applications, Safe offline RL, multi-agent RL.

Uncertainty Quantification: UQ for time-series forecasting models and generative models, conformal prediction

ML frameworks: **TensorFlow**, **Pytorch**, and **JAX**.

Data Science: AWS, **SQL, PostgreSQL**; distributed systems **Kubernectl**, **Hadoop on Spark**

Software Engineering: Competent in **Java** and **Python**, experience in **C, C++, R, Go, JavaScript/TypeScript**